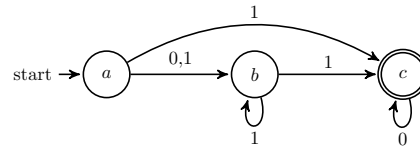


Test 3

1 Below is a the state diagram for an NFA N .



- a) For each string in the table below, indicate whether N accepts the string. For accepted strings, list (in order) a sequence of states the NFA goes through to accept the string. No other explanation needed.

string	accept? (Y or N)	sequence of states (for accepted strings)
01		
001		
101		
0101		
0110		

- b) Draw the state diagram for a DFA that accepts the same language using an algorithm that can convert any NFA into an equivalent DFA. (You need not show unreachable states.)

2 Let L be the language that contains all and only binary strings that contain a 1 and end with three or more consecutive 0's. (So $01000 \in L$ and $110000 \in L$, but $0100 \notin L$ and $000 \notin L$.)

a) Design a DFA that accepts the language L .

b) Construct a regular expression for L . (You do not have to use the algorithm that converts any DFA into a regular expression.)

3 The productions rules for a grammar G with start symbol S , terminals a , b , and c and non-terminals X , Y , and Z is given below:

$$\begin{array}{lcl} G : & S & \rightarrow XY \\ & X & \rightarrow a \mid YZ \\ & Y & \rightarrow b \mid ZX \\ & Z & \rightarrow c \mid XY \end{array}$$

- a)** Give either a derivation or a derivation tree (also called a parse tree) that shows that G generates $bacab$ (i.e., that $S \Rightarrow^* bacab$).

- b)** Is G a regular grammar? Explain.

- c)** Is G a context free grammar? Explain.

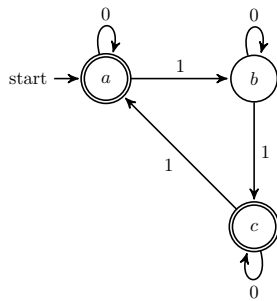
4 Explain how we know that $\{0^n 1^n \mid n = 0, 1, 2, \dots\}$ is not a regular language.

5 Show that if A is a regular language, then $\overline{A} = \{x \mid x \notin A\}$ is also regular.

6 We learned an algorithm for converting any DFA into a regular expression. This problem asks you a few questions about that algorithm.

a) In the first phase of the algorithm, we must modify the automaton to make sure it has two properties. What are those two properties?

b) Demonstrate how to do the first phase using the DFA below.



c) In the second phase, states are “ripped out” of a GNFA until only two states remain. Using that process, show what happens when state a is ripped out of the GNFA shown below. (You don’t have to complete the process, just do this one step.)

